



Computer Memory

01

Do you recognise all these devices?

You're right! All of these are **Storage Devices** that store data.

Storage Devices

Memory card!

Hard disk!

Pen drive!

Whatever we do in our day-to-day life is stored in our brain. Similarly, computers store all the tasks that we perform. Whether we create a document or edit an image, it requires some space to be stored. The computer's memory provides this space.

What is Memory?

In a computer, memory is the storage space where data is processed and information is stored. Hardware devices that contain the memory are called storage devices.

1. Computer Memory

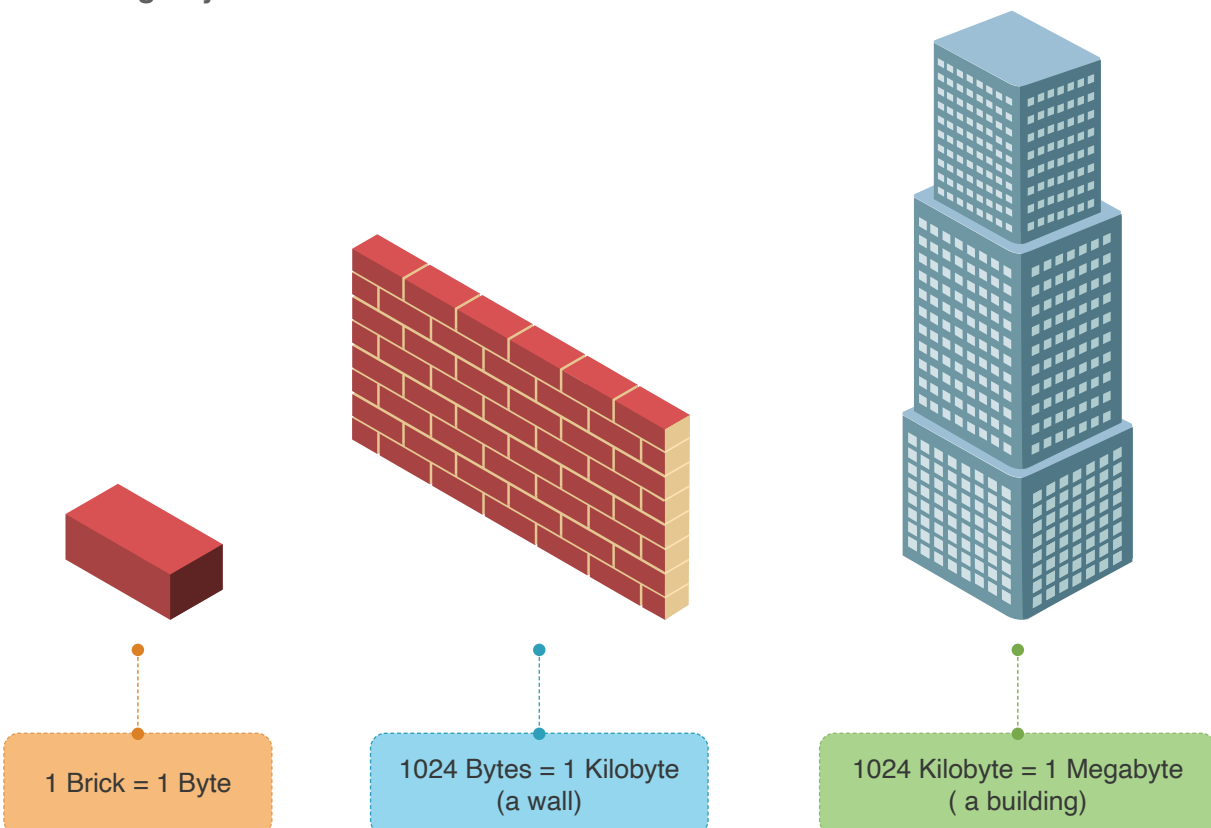
Basic unit of memory

Different quantities are measured in different units. Similarly, the quantity of memory occupied in a computer is measured in **bytes**.

Here is a table to remember memory conversions:

1 Byte	8 bits
1 Kilo Byte (KB)	1024 Bytes
1 Mega Byte (MB)	1024 Kilo Bytes
1 Giga Byte (GB)	1024 Mega Bytes
1 Tera Bytes (TB)	1024 Giga Bytes

For a better understanding of memory sizes, let's consider one brick to be 1 **Byte**. Then 1024 bricks will make one wall which we can call a **Kilo Byte**. Similarly, we can add 1024 walls to create a building which we can call a **Mega Byte**.



1. Computer Memory

Sizes of different files

- 1 Byte – Is a character like “A”
- 1 KB – Is about the size of a paragraph in Notepad
- 1 MB – Is about the size of a one-minute audio clip
- 1 GB – Is about the size of a one-hour HD video

Language of Computers

Computers do not communicate like humans. Computing language is different to the languages we speak. Computers use signals like “ON” and “OFF” to communicate.

- These signals are denoted by numbers.
- Computers only use two numbers to communicate: 0 and 1.
- ‘0’ is used to denote OFF and ‘1’ is used for ON.

When a computer wants to communicate, it sends a series of 0s and 1s. These 0s and 1s are sent in bundles of 8. Each digit is **one bit** and each bundle of 8 bits is a **Byte**.

Bytes

A unit is a standard that we use to measure quantities.

For example:

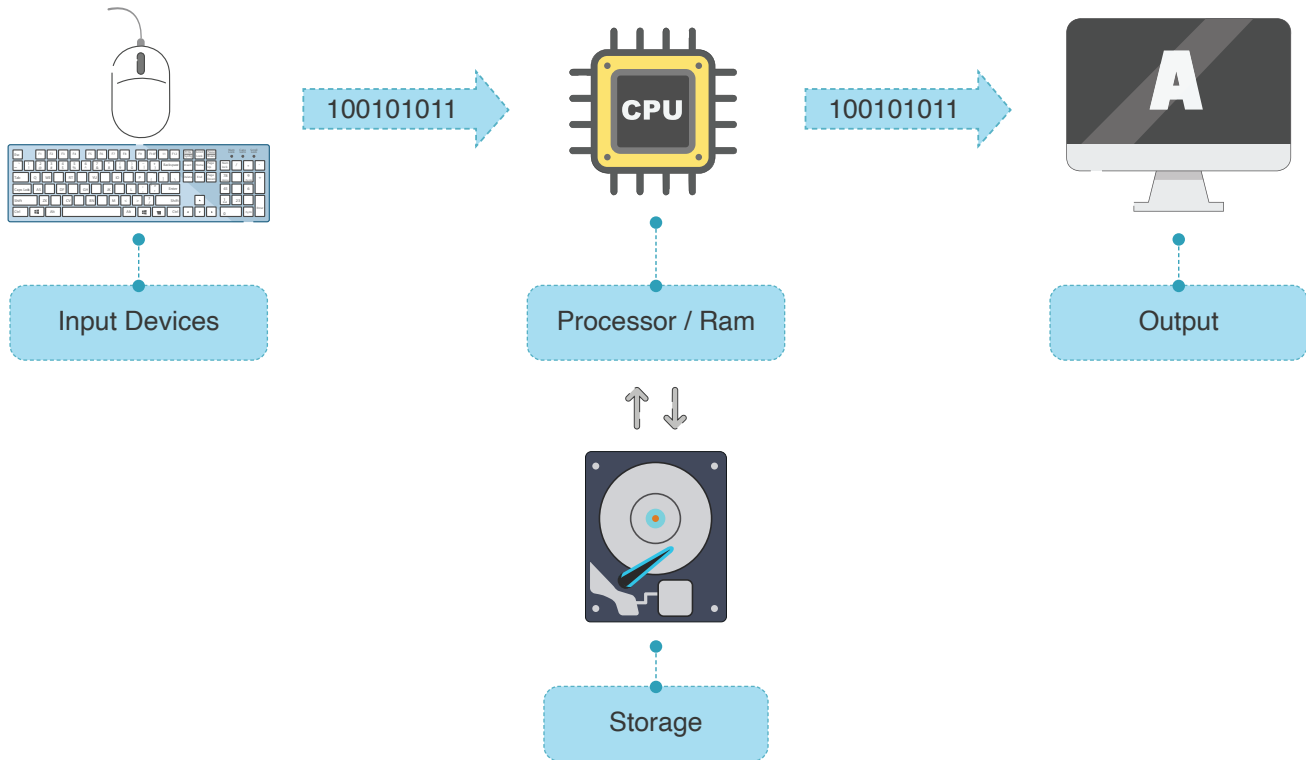
Units to measure mass: gram, kilogram, ton, etc.

Units to measure length: centimetre, metre, millimetre, etc.

Units to measure distance: metre, kilometre, etc

1. Computer Memory

The computer performs 4 simple operations:



1. The computer takes input from input devices like a keyboard, mouse, etc. in the form of 0s and 1s (bytes).
2. Data travels from input devices to the processor/RAM.
3. The processor sends the data to storage devices to keep it stored. After it is stored, memory devices send a copy of the data to the processor/RAM.
4. The processor sends the data to output devices like a monitor, speakers, etc.

Exercise

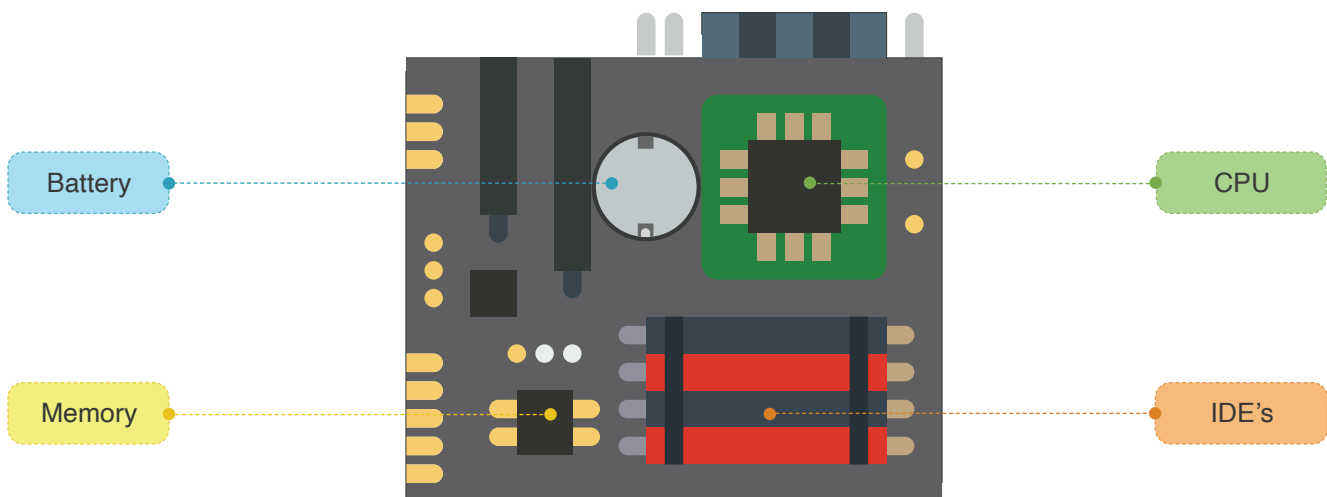
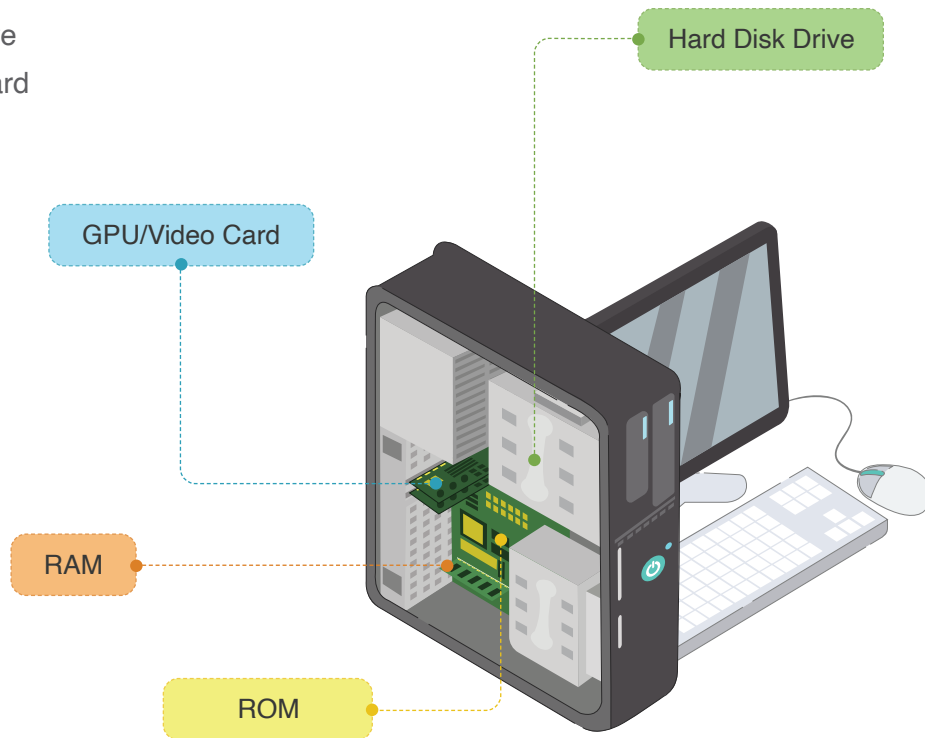
Fill in the blanks

- 1 The Unit of Memory is _____.
- 2 If 1KB is 1024 Bytes, then 3 KB is _____ bytes.

1. Computer Memory

Storage Devices inside a CPU Cabinet

- ROM
- RAM
- Hard Disk Drive
- GPU/Video Card



Motherboard Diagram

Data Storage Devices

In a computer, data is stored in a **Digital Storage Device** or **Memory Device**. This task is performed by computer's internal components. To manage these memory devices, the computer has a management system called the **Memory Management Unit (MMU)**.

The Memory Management Unit is a part of the CPU in the cabinet.

Data storage devices can be categorised into two main types -

- a. Temporary Memory Device
- b. Permanent Memory Device

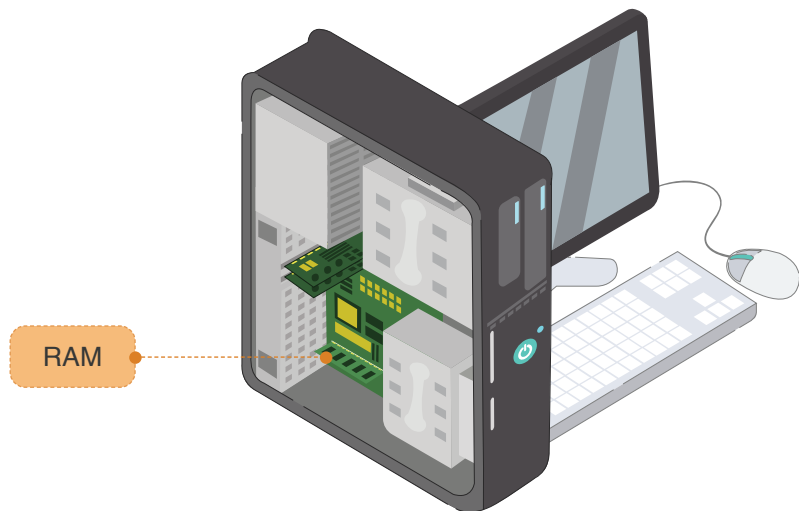
a. Temporary Memory Device

Memory devices that store information while a computer is switched on are called temporary memory devices. When it is switched off, the data from these devices is erased.

Examples of temporary storage devices:

- RAM
- Video Card/GPU

RAM (Random Access Memory)

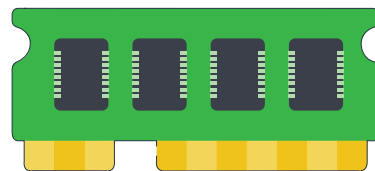


RAM is the primary memory of the computer. When a new program starts on the computer, its information is stored in the RAM.

- RAM stores data for a short period of time.
- It is connected to the motherboard inside the CPU cabinet.

For example:

When we open an image editing software, RAM accesses data related to the software and image.



Random Access Memory (RAM)

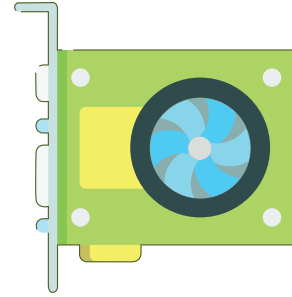
Video Card/GPU

Graphics Processing Unit (GPU) and RAM work similarly, but GPU works especially for high-quality videos and computer animations. Therefore, it is also called a Video Card.

For example:

When we play games in high definition, data is stored in the GPU instead of the RAM.

- A Video card/GPU is connected to the motherboard.
- GPU helps us play high definition video games and make computer animations and drawings.



Exercise

Fill in the blanks

- 3 Data in a computer is stored in _____.
- 4 The unit that manages storage devices in a computer is called _____.
- 5 The full form of RAM is _____.
- 6 The full form of GPU is _____.

Bytes

The first ever RAM was developed by Dr. Bernard Widrow in 1952-1953.

b. Permanent Storage Device

Memory devices that retain information even after a computer is switched off are called permanent storage devices.

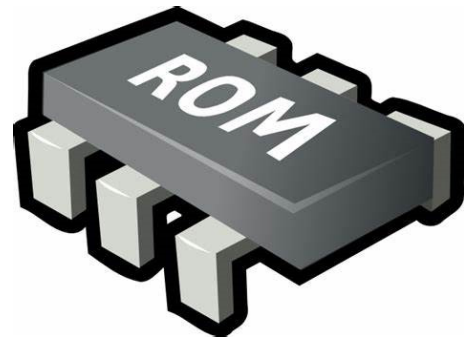
Examples of Permanent Storage devices:

- ROM
- Hard Disk or SSD

ROM (Read Only Memory)

ROM is the permanent memory that stores read only data. A user cannot save or remove data in ROM.

- ROM is located on the motherboard.
- ROM is used for a computer's start-up operations.
- Data stays stored in the ROM even after the computer is switched off.



Memory Operations

Read Memory - Read Memory is the process through which the CPU accesses the memory and processes information.

Example: An image saved on a computer that has to be shown on the monitor.

Write Memory - Write Memory is the process through which information is added or changed on a storage device.

Example: Changes made in a Word document.

Erase Memory - Erase Memory removes or deletes data that was stored in a device.

Example: To delete a file from the computer.

Permanent Storage Devices -

- a. Hard Disk Drive (HDD)
- b. Solid State Drive (SSD)

Hard Disk (HDD)

Hard Disk Drive (HDD) is a permanent storage device found in most computers. It stores the information, files, and folders that are on the computer.

A typical HDD used in modern day computers ranges from 512 GB to 4 TB in size.



Solid State Drive (SSD)

Solid State Drive (SSD) is an upgraded version of a Hard Disk. SSD is commonly used in modern day computers. Its average size ranges from 512 GB to 2TB.



Hard Disk Drive (HDD)	Solid State Drive (SSD)
The speed to read and write data is slower than SSD.	The speed to read and write data is faster than HDD.
HDD requires more power than SSD to perform tasks.	SSD requires less power than HDD to perform tasks.
Data may get erased over time.	Data doesn't get erased over time.

Activity

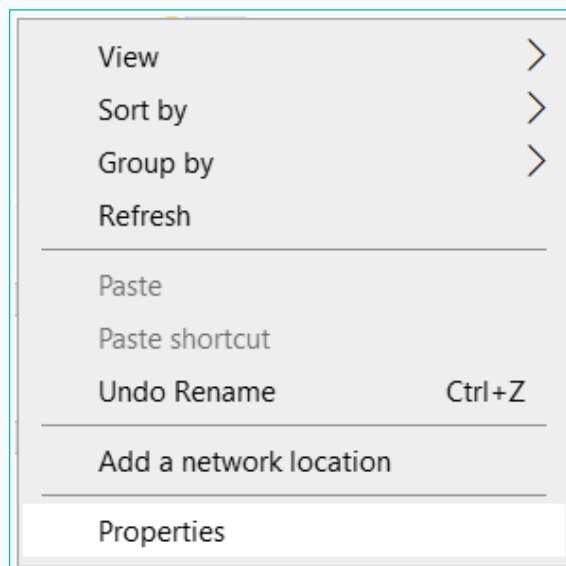
Check the memory specifications on a personal computer and note it down.

To check the memory of the computer:

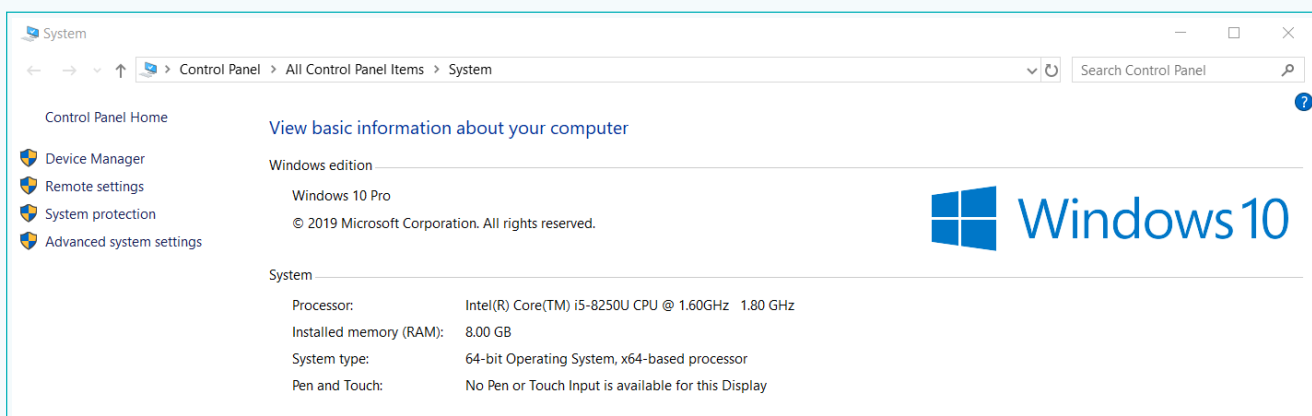
- Click on 'This PC'.
- The space available on the Hard Disk or Solid State Drive will be displayed.
- This is the ROM Memory.

To check RAM:

- Right click on the empty space on the Desktop.
- Click on 'Properties'.

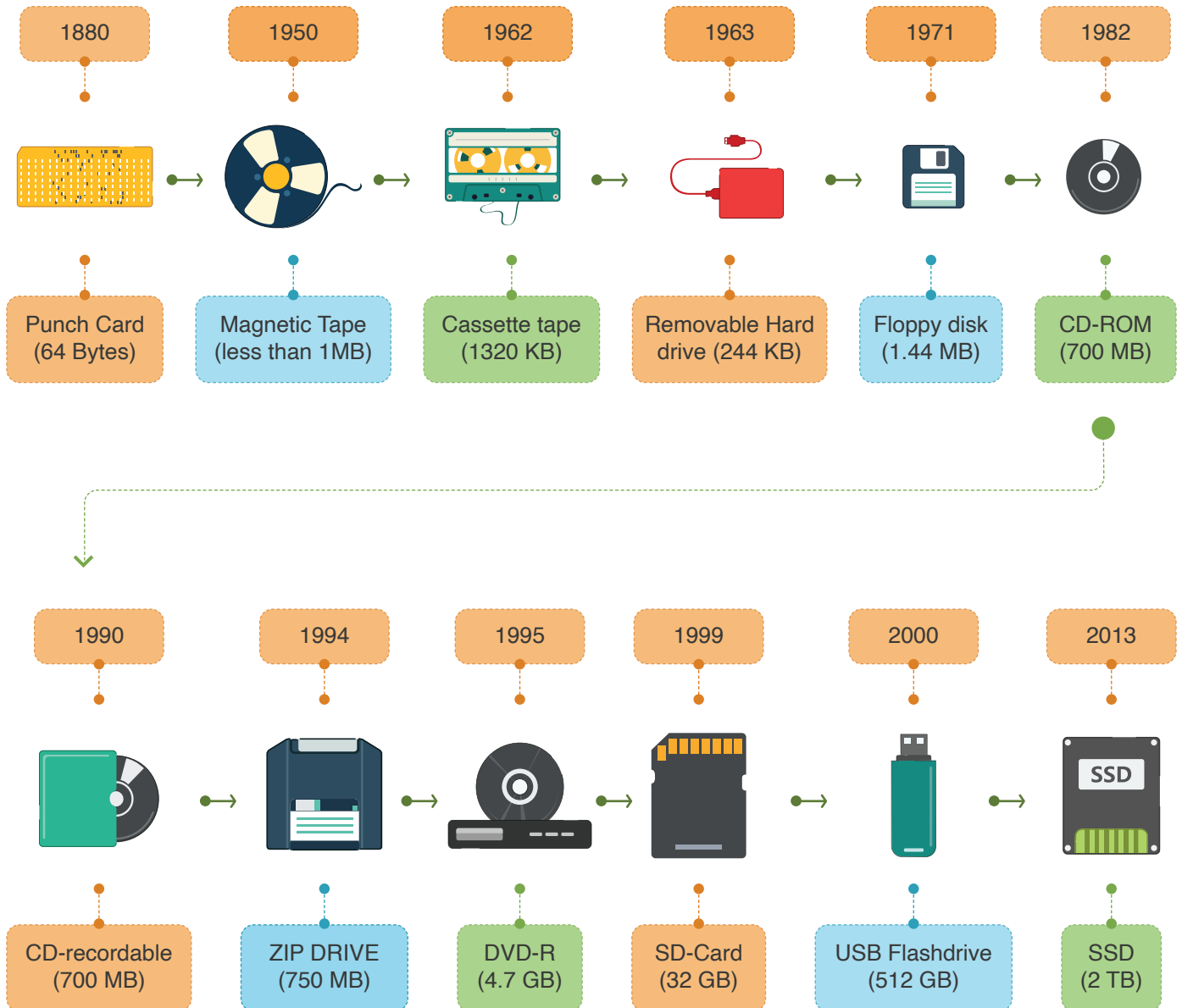


- Computer memory and its other specifications will be displayed.



1. Computer Memory

THE EVOLUTION OF DATA STORAGE



A computer contains different types of memory devices. All of them are responsible for temporary and permanent data storage. This data is used by the computer to operate and perform calculations.

Exercise

Fill in the blanks

- 7** 1 GB is equal to _____ MB.
- 8** When a program is loaded in the computer, its information is saved in _____.

State whether True or False

- 9** RAM is a permanent memory.
-  _____
- 10** SSD is faster than HDD.
-  _____
- 11** Video Card/GPU is used for gaming.

 _____

That's right Shreya. All storage devices have their capacities written on them.

Oh, this is what the 8GB written on my pen drive means.



Tech Flash

- **Memory** : Memory is information that is stored in the computer.
- **Byte** : Computers store information in a series of 1s and 0s. Each number is called a bit. 8 bits make one Byte.
- **Read Data** : Read Data is the process that reads information on the disk.
- **Write Data** : Write Data is the process that changes or saves new information on the disk.
- **RAM** : RAM (Random Access Memory) is used to load computer programs.
- **ROM** : ROM (Read Only Memory) is used to start a computer operation.
- **GPU** : GPU (Graphics Processing Unit) is similar to RAM. It helps to load high quality videos and computer animations.
- **HDD** : HDD (Hard Disk Drive) is used to save information in the computer.
- **SSD** : SSD (Solid State Drive) is an upgraded version of HDD and helps save information in the computer.